Game Theory for Data Science

Midterm Exam • Solutions and Grading Guide

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Answer summary. A1 F, A2 F, A3 T, A4 T, A5 T, A6 F. B1 (c), B2 (b), B3 (c), B4 (b), B5 (d), B6 (b).

Grading conventions. Part A: 1 pt for the correct T/F, 2 pts for a valid justification. A justification that is correct but supports the wrong letter earns 0. A correct letter with a vague justification ("by definition") earns 1 of the 2 justification points. Part B: 3 pts for the correct choice, 0 otherwise (no partial credit). Part C: see the rubric for each part.

Part A: True or False

A1. False. Matching pennies (payoffs ± 1) has no pure Nash equilibrium: in every pure profile one player wants to switch. Nash's theorem guarantees an equilibrium only in mixed strategies (here, both players randomize $\frac{1}{2}$–$\frac{1}{2}$).

Grading: full justification needs a specific counterexample or a precise statement that existence is only guaranteed in mixed strategies.

A2. False. Entry game: an entrant chooses Out (payoffs 0 to the entrant, 4 to the incumbent) or In; after In the incumbent chooses Fight $(-1, 0)$ or Accommodate $(2, 2)$. The profile (Out; Fight if In) is a Nash equilibrium, since the entrant expects -1 from entering and the incumbent's choice is off the path. It is not subgame perfect because Fight is not optimal in the subgame after In ($0 < 2$).

Grading: any valid non-credible-threat example earns full credit.

A3. True. In a finite two-player zero-sum game, (x, y) is a Nash equilibrium iff x is a maximin (optimal) strategy for player 1 and y is a minimax (optimal) strategy for player 2. Since x and y' are both optimal, (x, y') is an equilibrium, and it yields the value of the game (equilibria are interchangeable and equivalent).

Grading: 1 pt of justification for "same value"; full credit requires the link between equilibria and optimal (maximin/minimax) strategies.

A4. True. Under $p(i, j) = x_i y_j$, a row player told to play i (with $x_i > 0$) believes the column action is distributed according to y, independent of the recommendation. Since x is a best response to y, every i in the support of x is a best response to y, so obeying is optimal. The same holds for the column player. So every Nash equilibrium is a correlated equilibrium.

A5. True. If a^* maximizes Φ, then for any player i and any deviation a_i',

$$u_i(a_i', a_{-i}^*) - u_i(a^*) = \Phi(a_i', a_{-i}^*) - \Phi(a^*) \leq 0.$$

No unilateral deviation is profitable, so a^* is a pure Nash equilibrium. (This is also why every finite potential game has a pure equilibrium.)

A6. False. Shapley's (1964) 3×3 example: fictitious play cycles through the action profiles with runs that grow geometrically, and the empirical frequencies do not converge. Convergence is guaranteed only for special classes (two-player zero-sum games, generic 2×2 games, potential games).

Grading: naming Shapley's game or describing a non-convergent cycling example earns full credit; "FP only converges in zero-sum games" earns 1 of 2 justification points (the claim is too strong but the direction is right).

Part B: Multiple Choice

B1: (c) $3/5$**.** If the other team plays Stag with probability p, Stag pays $6p$ and Hare pays $4p + 3(1 - p) = 3 + p$. Indifference: $6p = 3 + p$, so $p = 3/5$.

B2: (b) I and II only.

- I: told Go, the other surely stops, and Go gives $1 > 0$; told Stop, the other surely goes, and Stop gives $0 > -10$. CE.
- II: told Go, the other surely stops: obey. Told Stop, the other goes or stops with probability $\frac{1}{2}$ each: Stop gives 0, Go gives $\frac{1}{2}(-10) + \frac{1}{2}(1) = -4.5$: obey. CE.
- III: told Go, the other goes or stops with probability $\frac{1}{2}$ each, so Go gives $-4.5 < 0$ (the payoff of Stop): disobey. Not a CE.

B3: (c) $O(\sqrt{T \ln n})$**.** With learning rate η, Hedge's regret is at most $\frac{\ln n}{\eta} + \frac{\eta T}{8}$; choosing $\eta = \sqrt{8 \ln n / T}$ gives $\sqrt{(T \ln n)/2}$. Average regret per round therefore vanishes at rate $\sqrt{\ln n / T}$.

B4: (b) $4/3$**.** Roughgarden–Tardos: for nonatomic routing with affine latencies the price of anarchy is at most $4/3$, attained by Pigou's network. ($5/2$ is the bound for *atomic* unweighted congestion games with affine costs.)

B5: (d) matching pennies. A 2×2 game has an exact potential iff the deviators' payoff changes around the 4-cycle $(1,1) \to (2,1) \to (2,2) \to (1,2) \to (1,1)$ sum to zero. In matching pennies each of the four deviations gains 2 for the deviator, so the sum is $8 \neq 0$ and Φ would have to increase forever. Game (a) is a symmetric 2×2 game (every such game has an exact potential), (b) has potential values $3, 1, 0, 3$ on $(U, L), (U, R), (D, L), (D, R)$, and (c) is a symmetric two-link congestion game.

B6: (b) $(R, (b, d))$**.** Player 1 gets 1 and would get 0 from L (because player 2 plays b after L), so R is optimal; player 2's on-path choice d is optimal ($3 > 2$). Hence it is a Nash equilibrium. It is not subgame perfect because b is not optimal after L ($0 < 1$). Option (a) is the SPE; in (c) player 2 would switch to d; in (d) player 1 would switch to R.

Part C: Problems

C1 (12 points)

(a) If the top link carried $x < 1$, its latency $x^2 < 1$ would be strictly lower than the bottom link's, and bottom drivers would switch. So in equilibrium $x = 1$: all traffic uses the top link, with latency $1^2 = 1$, equal to the bottom link's latency (so nobody gains by switching). Total travel time $= 1$.

Rubric (3 pts).

- 1 pt: equilibrium condition (used links have equal and minimal latency).
- 1 pt: $x = 1$. 1 pt: total cost 1.

(b) $C(x) = x \cdot x^2 + (1 - x) \cdot 1 = x^3 + 1 - x$. $C'(x) = 3x^2 - 1 = 0$ gives $x^* = 1/\sqrt{3} \approx 0.577$ ($C'' > 0$, so this is the minimum). Then

$$C(x^*) = \frac{1}{3\sqrt{3}} + 1 - \frac{1}{\sqrt{3}} = 1 - \frac{2}{3\sqrt{3}} \approx 0.615.$$

About 42% of the traffic should use the slow-but-uncongestible bottom link.

Rubric (4 pts).

- 1 pt: correct objective. 2 pts: $x^* = 1/\sqrt{3}$. 1 pt: $C^* \approx 0.615$.

(c) PoA $= \dfrac{1}{1 - 2/(3\sqrt{3})} \approx 1.626$. This is larger than Pigou's $4/3$ for $\ell_1(x) = x$: steeper (higher-degree) latency functions allow larger inefficiency (with $\ell_1(x) = x^d$ the PoA grows without bound as $d \to \infty$).

Rubric (2 pts).

- 1 pt: numeric PoA (accept 1.62–1.63). 1 pt: comparison with $4/3$.

(d) The marginal-cost toll equals the congestion externality $x\,\ell_1'(x) = 2x^2$, evaluated at x^*: $\tau = 2/3$. Check: at x^*, top cost $= \frac{1}{3} + \frac{2}{3} = 1 =$ bottom cost, so drivers are indifferent and the split x^* is an equilibrium. (This is the unique such toll, since the equilibrium condition is $(x^*)^2 + \tau = 1$.)

Rubric (3 pts).

- 1 pt: idea of charging the marginal externality (or equalizing costs). 1 pt: $\tau = 2/3$. 1 pt: verification.

C2 (12 points)

(a) By symmetry each route carries 1500 vans (check: $p/100 + 40 = 40 + (3000 - p)/100$ gives $p = 1500$). Travel time $= 15 + 40 = 55$ minutes.

Rubric (2 pts).

- 1 pt: split; 1 pt: 55 minutes.

(b) For *any* flow, each congestible edge carries at most 3000 vans, so its travel time is at most $30 < 40$. Hence

$$t(s \to A \to B \to t) = \tfrac{x_{sA}}{100} + \tfrac{x_{Bt}}{100} < \tfrac{x_{sA}}{100} + 40 = t(s \to A \to t),$$

and similarly it is strictly shorter than $s \to B \to t$. The zig-zag path is strictly best for every van whatever the others do, so in equilibrium all 3000 vans use it. The new travel time is $30 + 0 + 30 = 60$ minutes (total 180,000 van-minutes).

Rubric (4 pts).

- 2 pts: argument that the zig-zag path is always strictly shorter (checking only the all-zig-zag profile, i.e., that alternatives cost $70 > 60$, earns 1 of these 2 pts).
- 1 pt: all vans use it. 1 pt: 60 minutes.

(c) Let p, q, r be the flows on sAt, sBt and the zig-zag path, with $p + q + r = 3000$. Total travel time is

$$C = \frac{x_{sA}^2}{100} + \frac{x_{Bt}^2}{100} + 40(p + q), \qquad x_{sA} = p + r,\ x_{Bt} = q + r.$$

C is convex and symmetric, so take $p = q$ and $r = 3000 - 2p$; then $x_{sA} = x_{Bt} = 3000 - p$ and $C = \frac{2(3000-p)^2}{100} + 80p$. Setting $C'(p) = -\frac{4(3000-p)}{100} + 80 = 0$ gives $p = 1000$. So $p = q = r = 1000$, each congestible edge carries 2000 vans, and

$$C^* = 2 \cdot \frac{2000^2}{100} + 40 \cdot 2000 = 160{,}000 \text{ van-minutes} \quad (\text{average } 53\tfrac{1}{3} \text{ min}).$$

(Check: the marginal cost of every path is 80.) Price of anarchy $= 180{,}000/160{,}000 = 9/8$.

Rubric (4 pts).

- 1 pt: correct total-cost expression.
- 2 pts: optimal routing $p = q = r = 1000$.
- 1 pt: $C^* = 160{,}000$ and PoA $= 9/8$.

(d) It hurt in equilibrium: travel time rose from 55 to 60 minutes for every van (Braess's paradox). It helped the optimum: the planner's best total time fell from 165,000 (without the connector, where the equilibrium was already optimal) to 160,000. Selfish vans ignore the delay they impose on others on $s \to A$ and $B \to t$, so they overuse the connector.

Rubric (2 pts).

- 1 pt: equilibrium got worse ($55 \to 60$). 1 pt: optimum improved, with the externality explanation.